Can a Normal Universe Explain This Map?

Out-of-Distribution Detection for Weak Lensing by Misfit, Consistency and Outlier Exposure

Sachin Konan

Abstract

I detect weak-lensing convergence maps that the simulator did not produce. My starting observation is that normal maps differ mostly because they come from different universes: a detector that ignores the five generating parameters barely beats chance on my synthetic distortions, while the same statistics, told the true parameters, detect them well. I therefore ask of each test map whether *some* normal universe can explain it. Three tests answer this: the misfit left after fitting the parameters, the disagreement between parameters fitted from different views of the map, and a classifier trained to recognize distorted maps. Their calibrated scores are averaged. The method scores 0.7037 on the public leaderboard, up from 0.5253 for the pipeline I started from, and scores the 10,000 test maps in about five minutes on one GPU.

1 Introduction

A weak-lensing analysis reads cosmology off the matter distribution by comparing observed convergence maps with simulated ones. The comparison is only as good as the simulator. A map shaped by physics or systematics the simulator leaves out — an unmodeled baryonic effect, intrinsic alignments, a PSF residual — will still be fitted, and will quietly bias the inferred parameters. Phase 2 of the FAIR Universe challenge asks for the step that should come first: flag such maps.

The task is easy to state. The training set holds 25,856 simulated maps, each paired with the five parameters that generated it: $\theta = (\Omega_m, S_8, T_{\mathrm{AGN}}, f_0, \Delta z)$, covering cosmology, baryonic feedback and a photometric-redshift shift. The test set holds 10,000 maps without parameters. Some come from the same simulator, and some do not. I return one score per map, larger for maps more likely to be out of distribution (OoD).

The natural first attempt ignores θ: model the distribution of training maps, or of their summary statistics, and flag the test maps that fall where training maps are rare. This works poorly here, and the reason is instructive. Normal maps differ mostly because they come from different universes. Across the 101 training cosmologies, the cosmology alone explains a median 43% of the variance of each of my 466 summary statistics; the random matter field of each realization, baryons, redshift errors and shape noise explain the rest. The distortions I want to catch are small next to this spread (Figure 1). A distorted map stays inside the cloud of normal maps. A density over the 466 statistics with no knowledge of θ scores 0.01–0.13 on six of my synthetic distortions, close to chance (0.013). Told the true θ, the same statistics score 0.19–0.74 (Figure 3a).

The parameters are therefore not a nuisance; they are what makes the problem tractable. The right question for a test map is not whether it looks unusual, but

can some normal universe explain this map?

I do not know θ for a test map, so I fit it. This exposes a second difficulty: the fit is free to move to parameters that mimic the distortion. An IA-like distortion looks like a lower S_8, and a baryon swap moves the fitted feedback parameters. After fitting, detection of the IA-like distortion drops from 0.74 to 0.04 (Figure 3).

My detector is built around these two observations. A *misfit* test scores what the best-fitting normal universe still fails to explain. A *consistency* test fits θ separately from different views of the map — large and small scales, different statistics, different networks — and scores their disagreement: a distortion rarely

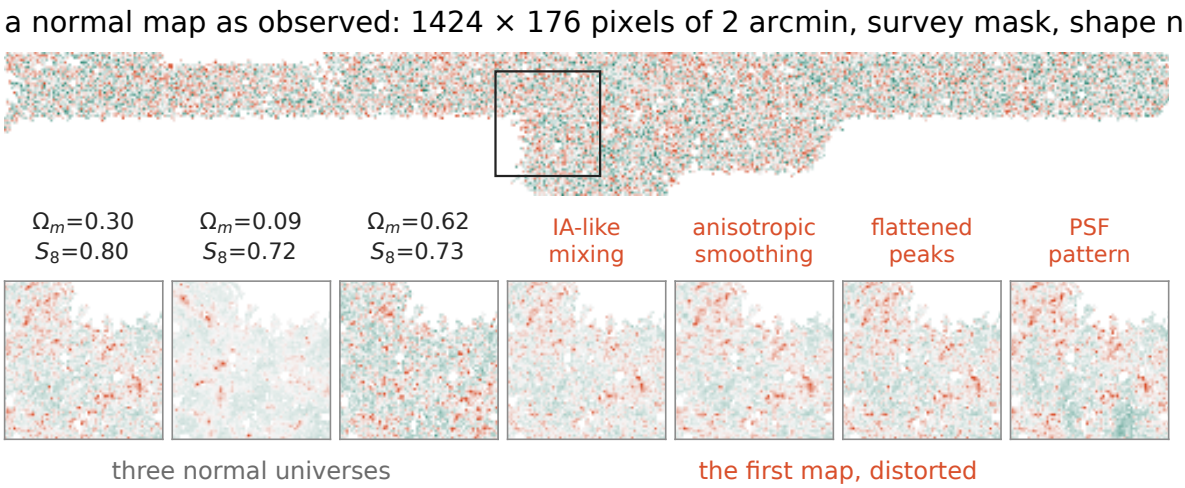


Figure 1: Normal universes differ more than distortions do. Top: a normal test-like map (survey mask, shape noise). Bottom: the same 128 × 128-pixel crop without noise, for three normal universes and for four of my synthetic distortions of the first. Changing Ω_m changes the map far more than any of the distortions, so a detector must judge a map against the universe that could have produced it.

pushes every view to the same wrong universe. An *outlier-exposure* test trains a network to recognize distorted maps directly, for distortions that fitting would absorb. The score is a weighted mean of the three kinds of calibrated z-scores (Figure 2). It reaches 0.7037 on the public leaderboard; each of the three tests moved it (Table 2).

2 Setup

Data. Each map covers 1424 × 176 pixels of 2 arcmin; 132,019 pixels lie inside the survey mask. I add Gaussian shape noise ($\sigma = 0.4/\sqrt{240}$ per pixel) to the clean training maps and round to float16, as in the test files. The 25,856 training maps are 101 cosmologies × 256 realizations, and each realization index carries the same baryon model ($T_{\rm AGN}, f_0$) in every cosmology. I therefore split at the level of maps, not realizations: *fit* (80%) trains every model, *calib* (10%) sets every normalization, and *audit* (10%) is held out for checks. A realization-level split would hold out whole baryon models.

Metric. The score is the true-positive rate averaged over 100 false-positive rates, log-spaced in $[0.001, 0.05]$. Since 59% of these points lie below 1%, the metric is decided by the few normal maps with the highest scores. Much of my design aims to keep that tail of normal scores light.

3 Method

3.1 Misfit: the best-fitting universe

For each family of features f I learn the distribution of normal maps given their parameters, $p(f \mid \theta)$. An MLP (3 layers of width 256) maps θ to a mean $\mu(\theta)$ and per-feature variances $\sigma^2(\theta)$, trained by maximum likelihood on fit maps. The residuals $u = (f - \mu)/\sigma$ are not Gaussian and are strongly correlated, so I model them with a Gaussian copula: each residual is mapped to a normal score through its empirical quantiles, and their correlation is estimated with OAS shrinkage and an eigenvalue floor of 0.02. The copula and the floor matter for the metric; without them a few normal maps reach very high misfit.

For a test map I fit θ by minimizing the negative log-likelihood, with a Gaussian prior on Δz ($\sigma = 0.022$) and a soft penalty for leaving the region spanned by the training cosmologies (Adam, 150 steps, three starts). The misfit is the copula energy at the fit, converted to a z-score with a location and scale that depend smoothly on $\hat{\theta}$, both estimated on calib maps.

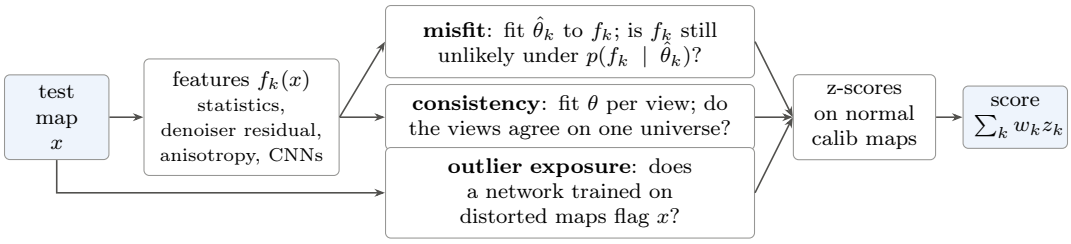


Figure 2: Three tests of one question: can some normal universe explain this map? Each test produces z-scores calibrated on held-out normal maps; the OoD score is their weighted mean. Every step treats one map at a time.

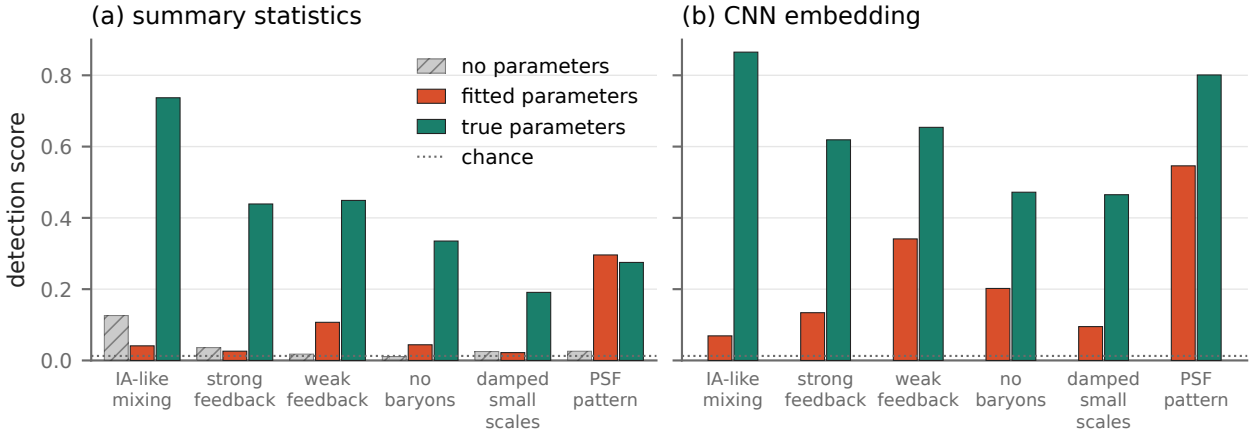


Figure 3: The parameters make distortions visible, and fitting them hides the distortions again. Detection of six synthetic distortions on held-out maps by the misfit of (a) the 466 summary statistics and (b) a CNN embedding. Without the parameters (gray, hatched) detection is near chance. With the true parameters (green) it is high. With fitted parameters (red), as at test time, most of the gain is lost: the fit moves to a universe that mimics the distortion. The consistency and outlier-exposure tests target this gap.

Five families serve as misfit members: difference-of-Gaussians wavelet norms (Sd); 112 statistics of the residual $x - D(x)$ of a U-Net denoiser D trained on normal maps (Dr), which isolate what the denoiser treats as noise; a noise battery of high-pass moments and tile variances (N2); the power spectrum with a binned bispectrum (PsBh); and power-spectrum anisotropy (X), the log power in wedges $|k_x| > |k_y|$ versus $|k_y| > |k_x|$ per scale band, aimed at PSF- and shear-like effects.

3.2 Consistency: do the views agree?

Fitting θ absorbs part of every distortion (Figure 3). But the fit to a distorted map is a compromise, and different views of the map compromise differently: a small-scale suppression pulls a fit to small-scale statistics but not one to large scales. I fit θ separately per view and measure how far the estimates spread, as a Mahalanobis distance of their deviations from the mean, with mean and covariance taken from calib maps. consP8 uses 8 views (two power-spectrum families, one-point PDF, extrema, Minkowski functionals, wavelet norms, and a CNN's embedding and its direct θ prediction). consRplus uses 13 views (four scale bands, two CNN embeddings and predictions, the denoiser families, and linear θ probes on the denoiser bottleneck and on the anisotropy features). xSL fits small- and large-scale statistics separately and scores how much worse each fits at the other's estimate.

3.3 Outlier exposure: has the map been distorted?

Some distortions are absorbed by every view, for instance an IA-like change that mimics a lower S_8 at all scales. For these I train a CNN to tell normal maps from distorted ones. Distortions are drawn from

member	test	weight	member	test	weight
Dr	misfit, denoiser residual	2.5	consRplus	consistency, 13 views	3
X	misfit, anisotropy	2	xSL	consistency, 2 scales	1
Sd	misfit, wavelet norms	0.5	consP8	consistency, 8 views	0.5
N2	misfit, noise battery	0.5	Ol	outlier exposure, logit	3
PsBh	misfit, spectrum + bispectrum	0.5	Oh	outlier exposure, misfit	1.5

Table 1: Members and weights of the final score. The consistency and outlier-exposure members carry 60% of the weight.

14 generator families at log-uniform strengths, including shear and PSF errors, redshift-distribution errors, wrong noise levels and masks, modified nonlinear and baryonic power, intrinsic alignments, source clustering and Poisson counts, and are applied to clean fit maps before noise. The network also predicts the generator family and θ, which shapes its features. Two members read it: Ol is its OoD logit as a robust z-score on calib maps, and Oh is the misfit of its outputs under the likelihood of the misfit test.

3.4 Combining the tests

Every member is a z-score with respect to normal calib maps, so the members share one scale. The final score is the weighted mean $s = \sum_k w_k z_k / \sum_k w_k$ (Table 1). The weights favor the consistency and outlier-exposure members; the plain misfit of the base statistics (Sd) carries 3% of the weight.

One map at a time. Every score is a function of its own map. Networks run in evaluation mode on fixed-shape zero-padded batches, and each map is fitted independently. All normalizations were fixed on training simulations; the test set is never used to fit, calibrate, normalize, rank or select anything. Leaderboard feedback was used only to choose among submitted weightings.

4 Validation

The test set's OoD maps come from distortions I do not know, so I validate on my own: 19 synthetic shifts of normal maps (shear bias, smoothing, misestimated and non-Gaussian noise, small-scale suppression and boost, flattened peaks, gradients, out-of-prior baryons, source clustering, Poisson counts, IA-like mixing, a PSF pattern). Validating on audit maps proved misleading: all their cosmologies are seen in training, and it rated as equal two weightings that the leaderboard separated by 0.08. The test maps' normal universes need not match the training ones, so I hold out whole cosmologies instead: my *leave-cosmology-out* proxy refits every member without 20 of the 101 cosmologies, in three disjoint sets, and scores the 19 shifts on the held-out maps.

The proxy ranked my main steps correctly (Figure 4b), but it is biased toward members built from the same ideas as my shifts. Raising the weight of the outlier-exposure logit from 3 to 6 improved the proxy on every fold and lowered the leaderboard by 0.004. I chose weights by the proxy and kept the leaderboard as the final judge.

5 Results

Table 2 lists every scored submission. The first rebuild of the likelihood alone barely moved the score (0.5253 to 0.5272): a better misfit test, by itself, still lets the fit absorb the distortions. The consistency test was the first large step, +0.040, and adding the denoiser residual with consistency at full weight gave the largest, +0.079. Outlier exposure gave the final step. Removing its misfit member Oh costs 0.020, and adding its logit Ol at weight 3 gains 0.036.

Interestingly, the leaderboard improved each time the plain misfit of the base statistics lost weight, from all of the score in v3-01 to 3% in the final weighting, and dropping the CNN-embedding misfit helped

submission	change	test added	public leaderboard
start	pipeline I started from	—	0.5253
v3-01	new conditional likelihood, map-level splits	misfit	0.5272 (+0.002)
v3-02	+ xSL, consP8	consistency	0.5669 (+0.040)
v3-04	+ denoiser residual; consistency at full weight	misfit, consistency	0.6461 (+0.079)
v3-15b	+ more views, noise, bispectrum, anisotropy, Oh	all three	0.6667 (+0.021)
v3-15d	− CNN-embedding misfit	—	0.6676 (+0.001)
v3-16b	+ Ol at weight 3 (final)	outlier exposure	**0.7037** (+0.036)
v3-15g	v3-15d without Oh	—	0.6479 (−0.020)
v3-16c	v3-16b with Ol at weight 6	—	0.6993 (−0.004)

Table 2: Every test moved the leaderboard. Public leaderboard score of each submission, with the change from the row above (the two rows below the line are compared with v3-15d and v3-16b). Adding consistency was worth 0.040 on its first try; the denoiser residual with consistency at full weight, 0.079; the outlier-exposure network, 0.056 in total (Oh 0.020, Ol 0.036).

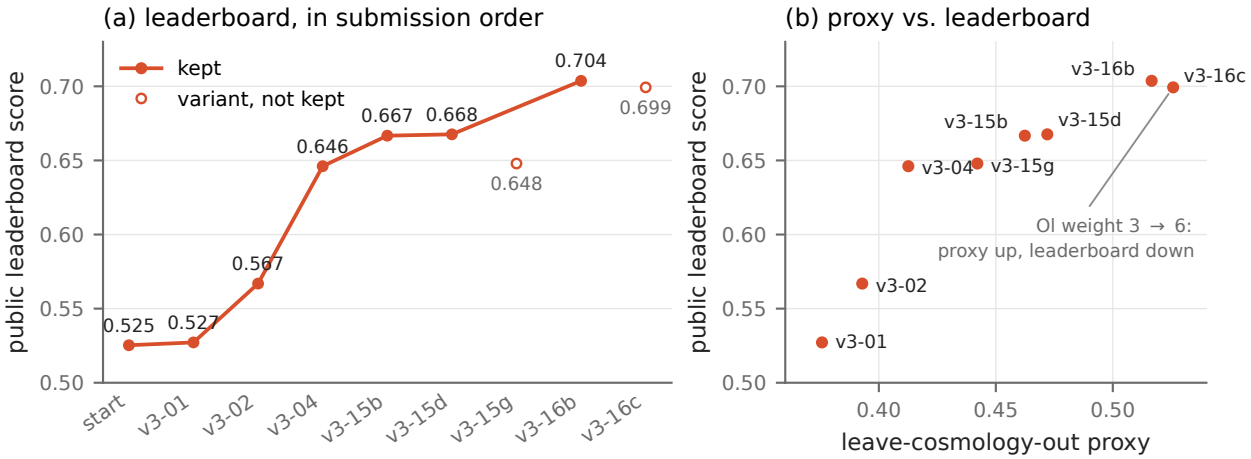


Figure 4: Progress on the public leaderboard, and how far my proxy predicted it. (a) Leaderboard score per submission; open markers are variants I did not keep. (b) The leave-cosmology-out proxy orders the main steps correctly but misses the last one.

slightly (+0.001). I read this as the test's normal maps sitting at universes my likelihoods fit less well than the training ones; tests that compare a map with itself, or with distortions, are less exposed to that.

Compute. Inference needs one GPU and PyTorch. Run with the organizers' ingestion program in their runtime image on one A100 with 4 CPU cores, the package scores the 10,000 public test maps in 306 s with a peak of 7.6 GiB of GPU memory, and reproduces my leaderboard scores (Spearman correlation 0.9997). Training is modest; the U-Net denoiser, for example, trains in 30 epochs on one A100.

6 Discussion

Three limitations are worth stating. First, a distortion that is exactly degenerate with the parameters — my IA-like mixing — passes both the misfit and the consistency tests; only outlier exposure sees it, and only when it resembles something the network was trained on. Second, my validation shares its assumptions with my detector: the synthetic shifts and the outlier-exposure generators come from the same list of plausible systematics, which is why the proxy overrated the members built on them. A validation set of distortion types withheld from all design choices would make the weights more trustworthy. Third, in my checks the highest-scoring normal maps come from universes at the edge of the training range (low Ω_m), where the

fitted feedback parameters often sit on the boundary and the outlier-exposure logit is also high. These maps set the low-FPR thresholds, so treating them better would lift the metric directly.

The broader lesson is about where to look. Normal maps vary mostly along the directions of the five parameters, and a detector that does not account for them sees mostly that variation. Conditioning on the parameters, and then guarding against the fit that conditioning requires, turned summary statistics that are near chance on their own into a detector that scores 0.70.